RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester)

Date	**11-12-2024**	**Time**	**10:00 AM-11:30 AM**	
Test	**Test-II**	**Maximum Marks**	**50**	
Course Title	**Fundamentals of Linear Algebra, Calculus & Differential Equations**		**Course Code**	**MA211TB**
Semester	**I**	**Programs**	**AS, CH, IM and ME**	

Sl. No.	**Answer all questions**	M	CO	BTL
1(a)	Show that the curves $r^n = a^n\cos(n\theta)$ and $r^n = b^n\sin(n\theta)$ intersect each other orthogonally.	5	1	2
1(b)	Find the slope of the tangent to the cardioid $r = a(1 + cos\theta)$ at the point $= \frac{\pi}{3}$.	5	2	2
2(a)	The curve that an idealized hanging chain or cable assumes under its own weight when supported only at its ends in a uniform gravitational field is a 'catenary' given by $y = c\cosh(\frac{x}{c})$, show that the curvature at any point on this curve varies inversely as square of ordinate at that point.	5	3	3
2(b)	The curve traced by a point on a circle as it rolls along a straight line without slipping is a cycloid $x = a(t - \sin t),\ y = a(1 - \cos t)$, show that the radius of curvature at any point is $-4a\sin(\frac{t}{2})$	5	2	3
3	For the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$, obtain the circle of curvature at the point $(\frac{a}{4}, \frac{a}{4})$.	10	4	3
4	Expand $y = \log(1 + e^x)$ as a Maclaurin series up to the terms containing x^4 and hence obtain the series for $\frac{e^x}{1+e^x}$.	10	3	2
5(a)	Solve: $y'' + 2y' + y = e^{2x}$	5	1	1
5(b)	The displacement $x(t)$ of a mass-spring system satisfies the differential equation $x''(t) + 9x(t) = 0$, with initial conditions $x(0) = 1, x'(0) = 1$. Obtain $x(t)$ at $t > 0$	5	3	3

BTL-Blooms Taxonomy Level, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	10	20	10	5	25	20		-	-

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